

Complex scaling approach to black-hole quasinormal modes and continuum response

Okuto Morikawa

RIKEN iTHEMS

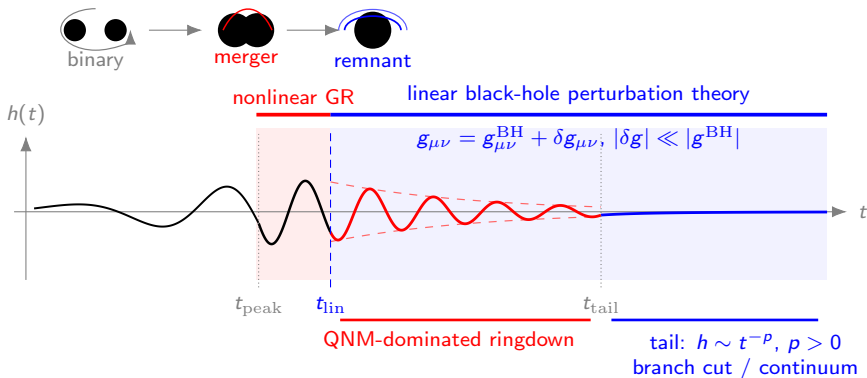
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Based on joint work with Shoya Ogawa and Takuya Hirose

- [PTEP [arXiv:2604.20442]], [arXiv:2605.03277], [arXiv:2608.14752].

Perturbation theory applies for $t \geq t_{\text{lin}}$

- **Open** black-hole system: ringdown and quasinormal modes (QNMs).

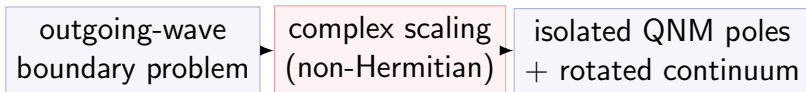


- QNM poles describe the exponentially damped, intermediate-time ringdown. [Price '72; Berti–Cardoso–Starinets '09]
- The late-time tail is still perturbative, but it is **not** QNM-dominated.

Why complex scaling for black-hole response?

- QNMs are **resonances**: poles of the analytically continued Green function.
- Continued fractions / exact WKB / time-domain methods can determine poles accurately. [Talks by Miyachi and Shiga, 15:45-]
- But the full response also contains a **non-pole continuum sector**.
- Question: can poles and continuum be represented by **one spectral problem**?

($\rightarrow$ universal class of resolvent singularities in hep / nucl / cond-mat / gr)



- Today: Schwarzschild/Reissner–Nordström $\rightarrow$ dS continuum response ($\rightarrow$ Riesz projector)

Black-hole perturbation as a 1D resonance problem

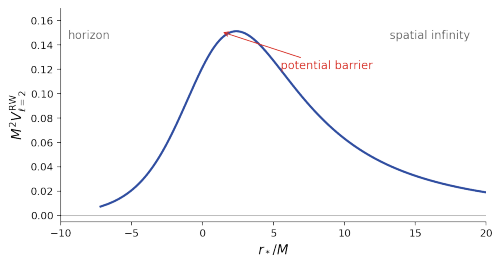
- Schrödinger-type master equation

$$\left[-\frac{d^2}{dr_*^2} + V_\ell(r) \right] \psi_\ell(r_*) = \omega^2 \psi_\ell(r_*).$$

- Radiation conditions

$$\psi_\ell \sim e^{-i\omega r_*} \quad (r_* \rightarrow -\infty), \quad \psi_\ell \sim e^{+i\omega r_*} \quad (r_* \rightarrow +\infty).$$

- The radiation conditions make the spectral problem non-self-adjoint.
- For Schwarzschild odd parity: **Regge–Wheeler barrier**.



Complex scaling regularizes outgoing waves

- Rotate the asymptotic coordinate into the complex plane:

$$r_* \longmapsto r_* e^{i\theta}, \quad 0 < \theta < \frac{\pi}{2}.$$

- For an outgoing wave,

$$\begin{aligned} e^{+i\omega r_* e^{i\theta}} &= e^{-(\operatorname{Re} \omega \sin \theta + \operatorname{Im} \omega \cos \theta) r_*} \\ &\quad \times e^{+i(\operatorname{Re} \omega \cos \theta - \operatorname{Im} \omega \sin \theta) r_*}. \end{aligned}$$

- It decays on the rotated contour when

$$\operatorname{Re} \omega \sin \theta + \operatorname{Im} \omega \cos \theta > 0.$$

- Resonance eigenvalues become discrete and θ -independent; the continuum rotates by 2θ in the $E = \omega^2$ plane.

outgoing boundary condition $\longrightarrow$ non-Hermitian L^2
eigenproblem

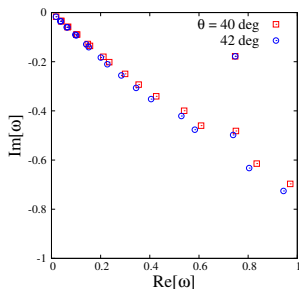
[Aguilar–Balslev–Combes theorem]

Poles separate from the rotated continuum

- Expand in a polynomial $\times$ real-range Gaussian basis,

$$\phi_{i,n}(r_*) \propto (\sqrt{\alpha_i} r_*)^n e^{-\alpha_i r_*^2},$$

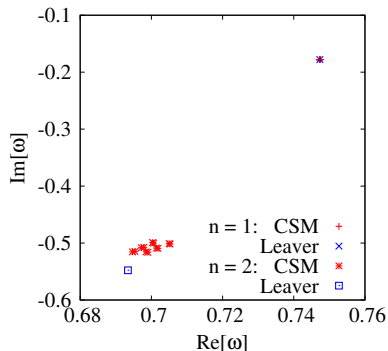
and solve a generalized non-Hermitian eigenvalue problem.



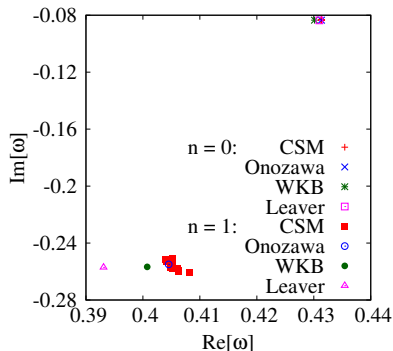
- **QNM candidate**: isolated and stable under changes of θ and basis parameters.
- **Continuum pseudostates**: move with the rotated cut.
- Pseudostates are quadrature data for the continuum, not extra QNMs.

CSM passes Schwarzschild and extremal-RN checks

- The finite-basis CSM reproduces narrow, isolated QNMs without explicit asymptotic Frobenius matching.



Schwarzschild, gravitational, $\ell = 2$



Extremal RN, odd gravitational, $\ell = 2$

- Fundamental Schwarzschild modes agree with Leaver to about 6–7 digits.
- Extremal RN remains accessible when the horizons coalesce and the naive Frobenius structure changes.
- Broad overtones near the rotated continuum are more basis-sensitive.

One resolvent contains poles and continuum

- The complex-scaled resolvent has the schematic spectral representation

$$R_\theta(z) = \sum_{n \in \text{res}} \frac{|\psi_{n,\theta}^R\rangle\langle\psi_{n,\theta}^L|}{z - E_n} + \int_{L_\theta} d\lambda \frac{|\psi_{\lambda,\theta}^R\rangle\langle\psi_{\lambda,\theta}^L|}{z - \lambda}.$$

- Pole positions set ringdown frequencies and damping rates.
- The non-pole sector gives prompt/nonresonant response and, in flat asymptotics, branch cuts and late-time tails.
- A finite L^2 basis replaces the continuum integral by pseudostates.

Keep the continuum as part of the response, not as numerical clutter.

CLD measures the continuum spectral shift

- For H and an asymptotic reference operator H_0 , define

$$\Delta\rho^\theta(E) = -\frac{1}{\pi} \operatorname{Im} \operatorname{Tr} \left[(E - H^\theta)^{-1} - (E - H_0^\theta)^{-1} \right].$$

- With $E = \omega^2$, the density plotted against frequency is

$$\Delta^\theta(\omega) = 2\omega \Delta\rho^\theta(\omega^2).$$

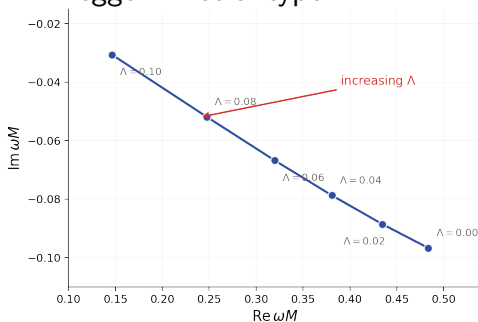
- Reference subtraction isolates the shift induced by the black-hole potential.
- Exact result: θ -independent. Residual dependence diagnoses finite-basis error.
- QNM spectrum: where are the poles? CLD: how is the continuum shifted?

Positive Λ lowers oscillation and damping scales

- Between the event and cosmological horizons, Schwarzschild–de Sitter has

$$f(r) = 1 - \frac{2M}{r} - \frac{\Lambda r^2}{3},$$

- Scalar, electromagnetic, and gravitational sectors remain Regge–Wheeler type.



Example: $s = 0$, $\ell = 2$, $M = 1$,

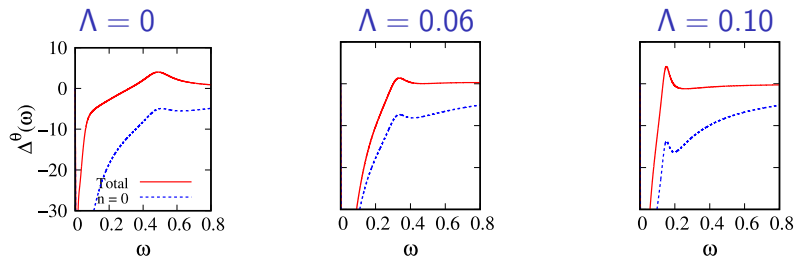
$$\Lambda = 0 : \quad \omega = 0.483646 \\ - 0.09676i,$$

$$\Lambda = 0.10 : \quad \omega = 0.146608 \\ - 0.030685i.$$

Lower frequency and slower damping toward the near-Nariai regime.

The CLD peak follows the QNM pole

- Scalar $s = 0$, $\ell = 2$; red is the total CLD, blue dashed is the lowest-pole partial reconstruction.



- Increasing Λ shifts the dominant structure to lower ω and sharpens it.
- The lowest pole controls the peak; higher poles and the continuum shape the smooth background and threshold behavior.

Pole dominance does not imply pole completeness.

CLD carries phase information, not the greybody modulus

- In ordinary one-dimensional scattering,

$$t(E) = |t(E)|e^{i\delta(E)}, \quad \Delta\rho(E) = \frac{1}{\pi} \frac{d\delta(E)}{dE}.$$

- The greybody factor is instead

$$\Gamma(E) = |t(E)|^2.$$

- **CLD**: trace-level spectral shift and transmission-phase information.
- **Greybody factor**: channel-resolved transmission probability.
- A resolvent trace cannot determine an off-diagonal channel amplitude or its modulus.

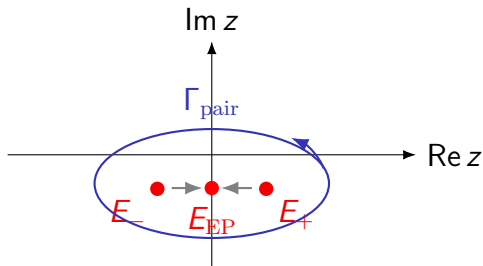
Next step: from $\text{Tr } R$ to $\langle g|R|f \rangle$.

Riesz clusters stay finite at an EP

- At a second-order EP, two QNM eigenpairs coalesce: separate residues become ill-conditioned while their combined response stays finite.
- Enclose the pair by one contour Γ_{pair} . Say $z = \omega^2$.
- Write $P = P_{\text{pair}}^\theta$ and $M = M_{\text{pair}}^\theta$:

$$P = \frac{1}{2\pi i} \oint_{\Gamma_{\text{pair}}} R_\theta(z) dz,$$

$$M = \frac{1}{2\pi i} \oint_{\Gamma_{\text{pair}}} z R_\theta(z) dz.$$



- P fixes the root space; M fixes the operator action inside it.

Two moments determine the pair resolvent

- Define the cluster center, traceless part, and squared half-splitting:

$$\begin{aligned}E_c &= \frac{1}{2} \text{Tr } M_{\text{pair}}^\theta, \\K_{\text{pair}}^\theta &= M_{\text{pair}}^\theta - E_c P_{\text{pair}}^\theta, \\ \Delta^2 &= \frac{1}{2} \text{Tr } [(K_{\text{pair}}^\theta)^2].\end{aligned}$$

- **Proposition:** The exact rank-two cluster resolvent is,

$$R_{\text{pair}}^\theta(z) = \frac{(z - E_c)P_{\text{pair}}^\theta + K_{\text{pair}}^\theta}{(z - E_c)^2 - \Delta^2}.$$

- ▶ No separate mode labeling, no normalized Jordan chain.
 - ▶ Channel matrix $\Rightarrow$ exact pair/rest decomposition of transmission.
- **Proposition:** At fixed real frequency, no EP branch singularity for transmission and the greybody factor. (cluster stays isolated and no pole reaches the physical axis.)

The double pole fixes polynomial ringdown

- At the EP, $\Delta^2 = 0$ and

$$R_{\text{pair}}^\theta(z) = \frac{K_{\text{pair}}^\theta}{(z - E_{\text{EP}})^2} + \frac{P_{\text{pair}}^\theta}{z - E_{\text{EP}}}.$$

- For a source f and observation functional g ,

$$\langle g | R(z) | f \rangle = \frac{C_{-2}[g, f]}{(z - E_{\text{EP}})^2} + \frac{C_{-1}[g, f]}{z - E_{\text{EP}}} + \text{regular terms}.$$

- ▶ Retarded Fourier transformation gives

$$\psi_{\text{EP}}(t) = \Theta(t)(D_0 + D_1 t)e^{-i\omega_{\text{EP}} t}.$$

- ▶ D_0 and D_1 are fixed source–observer coefficients, not independent fit parameters.
- ▶ If $C_{-2}[g, f] = 0$, the linear-in-time term is absent: a channel selection rule.

- **Proposition:** Jost (Wronskian) = Riesz–Laurent coefficients.

Summary

- Complex scaling converts black-hole radiation conditions into a controlled **non-Hermitian spectral problem**.
- Schwarzschild, RN, and Schwarzschild–dS results show that one spectrum organizes

isolated QNM poles + **rotated continuum**.

- ▶ CLD probes the continuum spectral shift beyond a list of QNM frequencies.
 - ▶ Applications: higher dimensions, and string-inspired black holes.
- Channel matrix elements of the full resolvent extend the same framework to greybody factors and sourced response.
- Riesz moments keep the response **finite and mode-label-free** through an exceptional point.

Take-home message

- Once P_{pair}^θ is available, we have essentially won: H^θ is already known, so $M_{\text{pair}}^\theta = H^\theta P_{\text{pair}}^\theta$ follows immediately, and the rest is rank-two algebra.
- The ingredients are well established: Complex scaling is a standard tool in nuclear-resonance theory, while contour projection is standard spectral numerical analysis. The new point is to combine them into an EP-safe, pair-resolved black-hole response.

Backup: finite-basis stability test

- Basis expansion:

$$\psi^\theta(r_*) = \sum_j c_j \phi_j(r_*).$$

- Generalized eigenvalue problem:

$$\sum_j \left(H_{ij}^\theta - \omega^2 N_{ij} \right) c_j = 0.$$

- Vary independently:
 - ▶ the scaling angle θ ;
 - ▶ the number, polynomial order, and geometric range of Gaussians;
 - ▶ the integration range and inverse tortoise map.
- Identify QNMs from isolated stationary branches; use the parameter-variation spread as a numerical stability estimate.
- Broad modes near the rotated continuum require better-balanced bases.

CLD reconstruction formulas

- For complex-scaled eigenvalues E_ν^θ and reference eigenvalues $E_{0,\nu'}^\theta$,

$$\Delta_{\text{tot}}^\theta(\omega) = -\frac{2\omega}{\pi} \text{Im} \left[\sum_\nu \frac{1}{\omega^2 - E_\nu^\theta} - \sum_{\nu'} \frac{1}{\omega^2 - E_{0,\nu'}^\theta} \right].$$

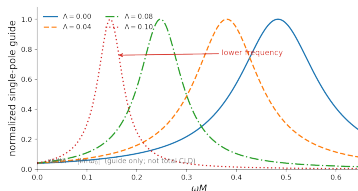
- Keeping only the lowest full-Hamiltonian pole gives

$$\Delta_{n=0}^\theta(\omega) = -\frac{2\omega}{\pi} \text{Im} \left[\frac{1}{\omega^2 - E_{n=0}^\theta} - \sum_{\nu'} \frac{1}{\omega^2 - E_{0,\nu'}^\theta} \right].$$

- The same reference contribution is subtracted in the total and partial curves.
- Near-threshold behavior also reflects reference subtraction and the Jacobian $dE/d\omega = 2\omega$.

Backup: Continuum response follows the same spectral reorganization

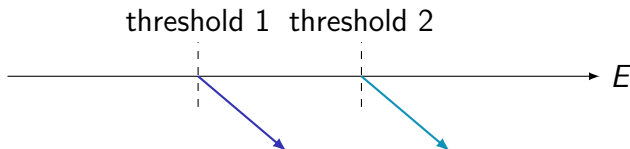
- In the full CSM CLD, the dominant structure **moves to lower frequency and sharpens** as Λ increases.



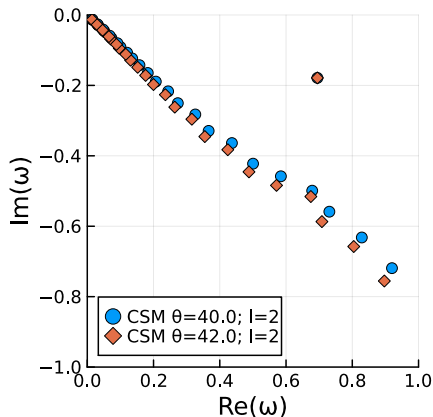
- The lowest-QNM partial reconstruction captures the main peak / shoulder.
- Higher poles + nonresonant continuum mainly modify the **smooth background** and threshold behavior.
pole dominance $\neq$ pole completeness

Backup: Beyond 4D single-channel problems

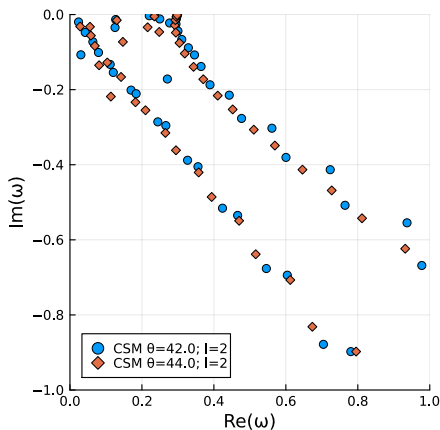
- Higher-dimensional dS
 - ▶ Tensor- and vector-type sectors remain Schrödinger-like.
 - ▶ Trial $d = 5$ calculations show stable low-lying resonances.
- String-inspired coupled channels
 - ▶ Matrix Schrödinger equations bring multiple asymptotic thresholds $\rightarrow$ multiple rotated continuum branches.
 - ▶ For the representative stringy 2D example, the present real-range Gaussian basis is not yet sufficiently stable.
 - ▶ Natural improvements: basis balancing, complex-range Gaussians, exterior complex scaling.



Backup: $d > 4$ and coupled channels



$d = 5$ Schwarzschild-dS, tensor sector



representative string-inspired 2×2 system

Backup: Riesz–Laurent operators

- At a second-order EP,

$$L_{-1} = P_{\text{pair}}^\theta, \quad L_{-2} = M_{\text{pair}}^\theta - E_{\text{EP}} P_{\text{pair}}^\theta.$$

- The source–observer coefficients are

$$C_{-1}[g, f] = \langle g | U_\theta^{-1} L_{-1} U_\theta | f \rangle,$$

$$C_{-2}[g, f] = \langle g | U_\theta^{-1} L_{-2} U_\theta | f \rangle.$$

- They are independent of Jordan-chain normalization and remain finite at coalescence.
- The double-pole coefficient factorizes in a compatible Jordan normalization:

$$C_{-2}[g, f] = \langle g | U_\theta^{-1} | \chi_0^R \rangle \langle \chi_0^L | U_\theta | f \rangle.$$

- Hence $C_{-2} = 0$ if the source misses the left EP state or the observation functional annihilates the right EP state.